

Zhiyu Wang

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Geometric Deep Learning

Large Language Model

AI4Biology/Healthcare



• Education

University of Cambridge

Advanced Computer Science, Master of Philosophy (Distinction, Rank: 8/60)

– Courses: Geometric Deep Learning, Affective AI, Mobile & Wearable ML, NLP, Multi-agent RL

• Cambridge, UK
• Oct 2024 — Jul 2025

University College London

Computer Science, Bachelor of Science (First Class Honors)

– Main Courses: Machine Learning, Computer Vision, Reinforcement Learning, Mathematics & Statistics, Software Engineering

• London, UK
• Sep 2021 — Jun 2024

• Research Experience

Protein Substructure Alignment via Optimal Transport | Shanghai Jiao Tong University

Research Assistant | Supervisor: Prof. Pietro Liò & Prof. Liang Hong

• Shanghai, China
• Apr 2025 — Sep 2025

- Developed PLASMA, a module based on optimal transport that can be seamlessly added to any protein representation learning model with minimal training or even no training to enable fast and highly accurate identification of locally similar substructures. ([Paper](#))
- Completed extensive experiments on 7 backbone protein language models across 3 tasks from the VenusX dataset (motifs, active sites, and binding sites), demonstrated consistent and significant improvement (+10-30% ROC-AUC) across all backbones and tasks.
- Conducted case studies using PyMol visualizations and proved that the founded similar substructures are indeed similar in structure and they are not found just because of sequence similarity.

Hierarchical Protein Representation Learning | University of Cambridge

Research Assistant | Supervisor: Prof. Pietro Liò

• Cambridge, UK
• Oct 2024 — Sep 2025

- Developed TCPNet, an SE(3)-equivariant topological neural network that simultaneously learns protein residue-level and secondary structure-level features without losing any geometric information. ([Invited Talk](#)) ([Paper](#))
- Awarded Highly Commended M.Phil Project Prize 2024-2025 by the Department of Computer Science and Technology
- Demonstrated 5% improvement on average across 3 popular protein representation learning tasks when only structure features are provided, which can be particularly important for de novo antibody design.

Multi-Omics Integration Data with GNNs | University of Cambridge

Research Assistant | Supervisor: Prof. Pietro Liò

• Cambridge, UK
• Feb 2025 — Sep 2025

- Using heterogeneous graph network for embedding single cell multi-omics (e.g. RNA, protein, and ATAC modalities) data to a shared latent space and conduct cell clustering. ([Paper](#))
- Training time on BM-CITE and LUNG-CITE datasets takes less than an hour on a laptop GPU, with 3% lower ARI and 2% lower NMI compared to the statistical method MOJITOO.

Graph-based Pain Estimation from Facial Expressions | University of Cambridge

Research Assistant | Supervisor: Prof. Hatice Gunes

• Cambridge, UK
• Oct 2024 — Jan 2025

- Presented at IJCAI25 MiGA Workshop. ([Paper](#))
- Proposed the GraphAU-Pain model that fuses a ResNet-50 backbone with graph-based facial action unit (AU) representation learning for pain intensity estimation, allowing better clinical interpretability and generalization compared to purely image-driven methods.
- Developed a novel transfer learning strategy to address domain differences and missing annotations, improving GraphAU-Pain's AU occurrence prediction F1-score by 30% on the UNBC dataset.

Readmission Prediction with Remote Patient Monitoring Data | University College London
Research Assistant | Supervisor: Prof. Ivana Drobnjak

• London, UK
• May 2023 — Jun 2024

- Collaborated with clinicians from UCL Hospital to develop a SQL-based data pipeline for processing raw remote patient monitoring data, performing clinical feature engineering, and building interpretable ML models (ROC-AUC 0.79 ± 0.03) with SHAP analysis.
- Designed a patient-independent stratified bootstrapping evaluation framework to address extreme data scarcity (55 patients) and class imbalance, preventing data leakage and ensuring robust generalization estimates for clinical deployment.

• Industrial Experience

Hierarchical Reasoning Model for Healthcare | Sapient Intelligence
AI Research & Application Engineer

• Beijing, China
• Sep 2025 — Present

- Applying **Hierarchical Reasoning Models** to healthcare scenarios and collaborating with Peking Union Medical College Hospital to develop healthcare solutions.

Healthcare LLM Knowledge Distillation | Xunfei Healthcare Technology
Assistant AI Research Engineer

• Hefei, China
• Jul 2024 — Sep 2024

- Built a knowledge distillation pipeline using a teacher LLM to synthesize medical instruction examples from human-annotated samples, improving student model (**Xiaoyi**) instruction-following accuracy from 34% to 82% via supervised fine-tuning.

RAG-Based Financial Intelligence System | Luojin Data Information
Intern Machine Learning Engineer

• Shanghai, China
• May 2023 — Aug 2023

- Developed a Retrieval-Augmented Generation (RAG) pipeline with MongoDB vector store and LangChain agents for automated financial report classification, achieving 92% query translation accuracy and 97% entity matching rate through semantic search.

Low-cost AI Chatbot Generation System for Hospitals | National Health Service
Intern Software Engineer

• London, UK
• Sep 2022 — May 2023

- Created an automated chatbot generation service for hospitals, reducing development time from weeks to 30 minutes.
- Delivered a live demonstration at Great Ormond Street Hospital to clinicians, and presented at UCL to IBM, Microsoft, and Intel, earning **media recognition** from an IBM Master Inventor.

• Additional Experience

Fresher Coding Sessions | University College London
Programming Tutor

• London, UK
• Sep 2022 — May 2023

- Organized biweekly group discussions for 12 first-year computer science students, offering academic guidance and peer support.